

Task 1: Basic Network Sniffer

Description

This project is a basic network sniffer developed using Python and the Scapy library.

The purpose of this project is to capture and analyze network packets to understand how data travels through a network. It displays important packet information such as source IP address, destination IP address, network protocol, packet size, and payload data.

Features

- Captures network packets
- Displays source and destination IP addresses
- Identifies network protocols (TCP/UDP)
- Shows packet length
- Analyzes packet payload data
- Helps understand network communication

Technologies Used

- Python
- Scapy Library
- Networking Concepts

How It Works

The program listens to network traffic and captures packets passing through the network interface.

For each packet, it extracts:

- Source IP → Sender address
- Destination IP → Receiver address
- Protocol → TCP/UDP communication type
- Payload → Data inside the packet

Installation

Install required library:

```
pip install scapy
```

Run Project

```
python network_sniffer.py
```

Learning Outcome

Through this project, I learned about:

- Packet analysis
- IP addressing
- TCP and UDP protocols
- Network traffic monitoring
- Basic cybersecurity concepts

Project Type

Cyber Security Internship Task - CodeAlpha

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IP(src="192.168.1.10", dst="1.1.1.1")/TCP()/Raw(load="Hello")

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for packet in packets:
 print("\n==== PACKET ====")

 print("Source IP:", packet.src)
 print("Destination IP:", packet.dst)

 if packet.haslayer(TCP):
 print("Protocol: TCP")

 elif packet.haslayer(UDP):
 print("Protocol: UDP")

 print("Size:", len(packet))

 if packet.haslayer(Raw):
 print("Payload:", packet[Raw].load)

▼ ... Requirement already satisfied: scapy in /usr/local/lib/python3.12/dist-packages (2.7.0)

==== PACKET ====
Source IP: 192.168.1.5
Destination IP: 8.8.8.8

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print("Protocol: UDP")

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print("Size:", len(packet))

if packet.haslayer(Raw):
 print("Payload:", packet[Raw].load)

▼ ... Requirement already satisfied: scapy in /usr/local/lib/python3.12/dist-packages (2.7.0)

==== PACKET ====
Source IP: 192.168.1.5
Destination IP: 8.8.8.8
Protocol: TCP
Size: 40

==== PACKET ====
Source IP: 10.0.0.2
Destination IP: 142.250.190.14
Protocol: UDP
Size: 28

==== PACKET ====
Source IP: 192.168.1.10
Destination IP: 1.1.1.1
Protocol: TCP
Size: 45
Payload: b'Hello'